Peer Review: MoveSync Motion Sensor System

Project Title: *A Low-Cost and Low-Complexity Motion Sensor System Using IMU and Video-Based Tracking*
Authors: Jack Hutton, Alexander Jonsson, Sahand Khodabandehloo

1. Goals and Scope

The report does a great job explaining why this project matters. It aims to take high-tech movement analysis out of the lab and make it accessible to coaches and athletes using only a web browser. The goals, such as tracking body joints with video and synchronizing that with sensor data, are clear and easy to follow.

How to go deeper:

While the goals are clear, the report should explain more clearly how much error is considered acceptable for practical sports analysis. For example, if the system estimates that an athlete ran 10 meters when the real distance was 8 meters, is that accurate enough to still be useful for coaches or athletes? Connecting the technical results to real-world requirements would strengthen the overall discussion. It would also help if the authors clearly defined the intended use case. For example, is MoveSync meant mainly for rough training feedback, comparison between sessions, or accurate biomechanical measurement? This would make it easier to judge whether the current accuracy is acceptable.

2. Background Knowledge

The background section is one of the strongest parts of the report. It explains IMU sensors, sensor fusion, and why the MoveNet Thunder model was selected for better accuracy in video analysis.

How to go deeper:

- **Magnetic Interference:**
  The report explains that magnetometers can be affected by hard-iron and soft-iron distortions, but it does not discuss in enough detail how these disturbances may affect the system in real sports environments. Since gyms and indoor training facilities often contain metal structures and equipment, it would be useful to explain how the system handles magnetic disturbances during practical use.
- **Choosing Filter Frequencies:**
  The report states that most human movement occurs below 10 Hz, but later uses a 20 Hz cutoff frequency in the filtering stage. A clearer explanation of this choice would improve the

discussion. For example, the authors could explain whether the higher cutoff was selected to preserve fast impacts or rapid sports movements.

- Since the authors asked what could be reduced, some general background explanations could be shortened. For example, the explanations of IMUs, Euler angles, quaternions, and sensor fusion are useful, but some parts read like textbook material. The report could use less space on general theory and more space on explaining the project's own results and limitations.
- A simple figure showing the full processing pipeline would improve clarity. For example, the figure could show video input and IMU input going through synchronization, filtering, sensor fusion, gravity removal, and then final sports metrics. A small diagram explaining gravity removal could also help because it seems closely connected to the distance errors.

3. How the System Works (Methods)

The system architecture is well-organized and easy to understand. Using a session-based structure to connect video recordings and IMU data is a strong design choice and makes the workflow easier to manage.

How to go deeper:

- **Testing Parameter Choices:**
  Some parameter choices seem not fully justified. To improve the scientific quality of the report, the authors should explain how these parameter choices affect the final results. For example, what happens to the estimated speed or distance if the filter settings are changed slightly? A small sensitivity analysis would strengthen the methodology section.
- **Choosing the Best Filtering Method:**
  The results show that the VQF method performs much better than the Madgwick filter for distance estimation. Because of this, the report should explain more clearly why Madgwick was initially selected and whether VQF is expected to become the main method in the final system.

4. Results and Problems

Using a synthetic (computer-generated) dataset was a very good decision because it allowed the authors to compare the results against known ground truth values.

How to go deeper:

- **Large Distance Errors:**
  The results show that the system overestimates distance by a very large amount, sometimes between 7 and 44 times the true distance. Instead of only presenting these errors, the report should analyze the possible causes more deeply. For example, are the errors mainly caused by sensor drift, inaccurate gravity removal, or problems with orientation estimation?
- **Physically Impossible Angles:**
  The report honestly mentions that some joint angles became physically unrealistic. To strengthen the analysis, the authors should discuss possible reasons for this problem. One

possible explanation could be small alignment errors between the sensor and the body segment, or synchronization errors between the video and IMU data.

5. Final Advice

- **Real-World Validation:**
  It would improve the report if the authors performed at least one simple real-world validation test, such as measuring a known walking distance, to compare real movement data against the synthetic dataset results.
- **Focus More on "Why":**
  The Discussion section should spend more time explaining why the remaining errors occur. A strong engineering report does not need perfect results, but it should show a clear understanding of where the limitations and technical problems come from.
- **Synchronization Accuracy:**
  Since synchronization currently depends on manual alignment, the report should discuss how sensitive the system is to small timing errors. For example, how much would a delay of half a second affect the calculated speed, distance, or motion metrics?
- Overall, the report presents a strong and useful prototype, but the main improvement should be deeper analysis rather than more background theory. The authors should focus on explaining why the errors happen, how sensitive the results are to synchronization and filtering choices, and what level of accuracy would be needed for practical sports use.